Unit: 6 Unit 5: Linear Equations
Topic: Scatter Plot and The Best Fit Line

Name: ________________
Date: ____________

Quiz (Habanero)

Q1. A researcher records the temperature t (in °C) and the number of birds b seen in a park. If the data points are $(25, 15)$, $(28, 20)$, $(30, 22)$, and $(32, 25)$, what type of correlation exists between t and b?

(A) Positive
(B) Negative
(C) None
(D) Cannot be determined
(E) Inverse

Q2. Which of the following best describes a scatter plot with the equation of the line of best fit $y = 2x - 3$?

(A) A vertical line with a y-intercept of -3
(B) A horizontal line with a y-intercept of -3
(C) A line with a slope of 2 and a y-intercept of -3
(D) A line with a slope of -3 and a y-intercept of 2
(E) A parabola with a y-intercept of -3

Q3. A farmer finds that the yield y (in kg) of wheat is related to the amount of rainfall r (in mm) by the equation $y = 0.5r + 200$. What yield can be expected if the rainfall is 600 mm?

(A) 350 kg
(B) 400 kg
(C) 500 kg
(D) 450 kg
(E) 550 kg

Q4. What is the purpose of drawing a line of best fit on a scatter plot?

(A) To find the y-intercept
(B) To identify the slope
(C) To make predictions based on the data trend
(D) To find the equation of the line
(E) To determine the type of correlation

Q5. A city's temperature t (in °C) and the amount of rainfall r (in mm) are recorded over several months. The data points are $(18, 20)$, $(22, 30)$, $(25, 35)$, and $(28, 40)$. What can be inferred about the relationship between t and r?

(A) As temperature increases, rainfall decreases
(B) As temperature increases, rainfall increases
(C) There is no correlation between temperature and rainfall
(D) The relationship is inverse
(E) The data is insufficient to make a conclusion

Q6. The equation of the line of best fit for a set of data is $y = -0.8x + 10$. What does the slope represent?

(A) The rate of decrease of y with respect to x
(B) The rate of increase of y with respect to x
(C) The y-intercept
(D) The x-intercept
(E) The correlation coefficient

Q7. A researcher collects data on the amount of sunlight s (in hours) and the growth rate g (in cm) of plants. The data points are $(4, 5)$, $(6, 8)$, $(8, 10)$, and $(10, 12)$. What is the equation of the line of best fit?

(A) $y = x + 1$
(B) $y = x - 1$
(C) $y = 0.5x + 3$
(D) $y = 1.5x - 2$
(E) $y = 2x - 3$

Q8. The line of best fit for a set of data has the equation $y = 0.4x - 2$. If $x = 15$, what is the predicted value of y?

(A) 2
(B) 4
(C) 6
(D) 3
(E) 5

Open Ended Questions (FRQ)

Q9. A meteorologist records the temperature t (in °C) and the amount of cloud cover c (in

Q10. A scientist studies the relationship between the amount of rainfall r (in mm) and the number of trees t in a forest. The data points are $(100, 50)$, $(120, 60)$, $(150, 70)$, and $(180, 80)$. Create a scatter plot and determine the type of correlation between r and t. Use the data to make a prediction about the number of trees when the rainfall is 200 mm.

Chef's Tips

- Ensure students have graph paper and calculators for the quiz
- Remind students to show all work and explain their reasoning for free response questions
- Allow students to use a dictionary or glossary for vocabulary-related questions